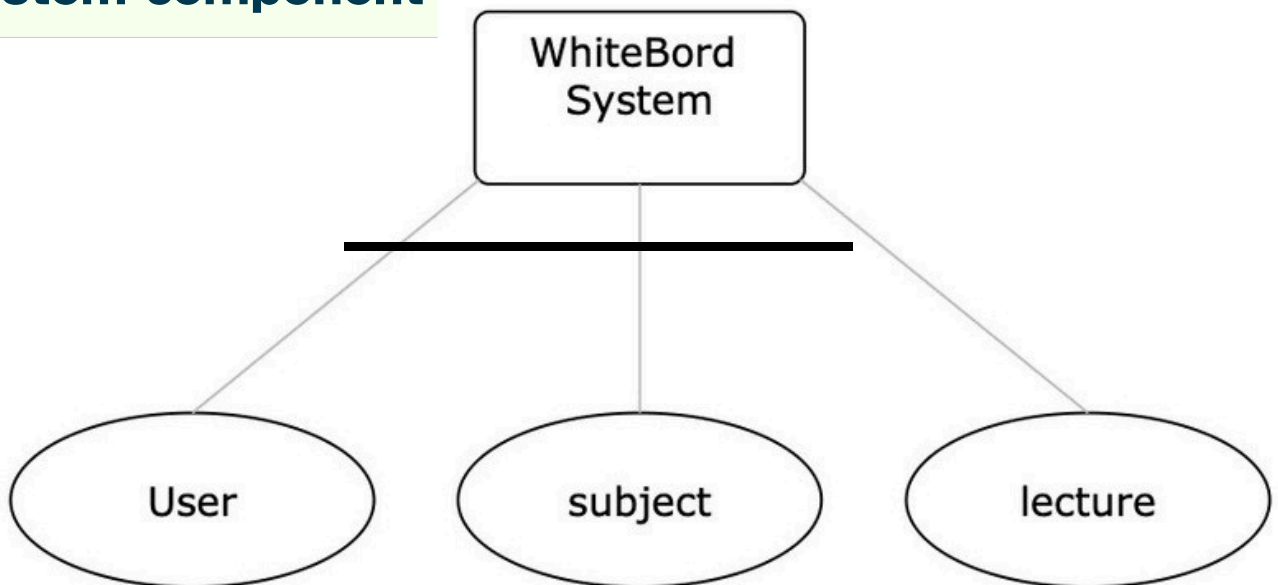


WHITEBOARD SYSTEM

- **pref**

Distance Learning System is a system designed to manage and facilitate the online learning process. It allows students to attend online lectures, access educational materials, submit homework and assignments, take online tests, and track their academic progress. It also helps instructors manage lectures, assignments, tests, and student performance.

- **system component**



1. **User**

ID
name
email
listof subjects
username
password
role

2. **subject**

subject_ID
name
descriotion
teacher
list of students

3. **lecture**

title
description
duration
date
list of
material

VALIDATION CHECKLIST

1.User

- ID
 - can not be empty
 - length smust be more than
- name
 - can notbe empty
 - the length between 4 and 15 letter
- email
 - must be valid email
- list of subjects
 - can not be Null
- username
 - can notbe empty
 - the length between 4 and 20
- password
 - cant be empty
 - the length between 8 and 20
 - have special leter
 - have number
 - have capital letter
- role
 - must be ether "Student"or" Teacher"

2.Subject

- subject_ID
 - can not be empty
 - length must be more than 2
- name
 - can not be empty
 - the length between 10and 25 letter
- description
 - can not be empty
 - the length more than 25
- tacher
 - can not be Null
- list of student
 - can not be Null

VALIDATION CHECKLIST

3. lecture

- title
 - can not be empty
 - the length between 4 and 18
- description
 - can not be empty
 - the length more than 25
- duration
 - cat not be null
 - the minmum duration 30 and max 90 menites
- date
 - should be a date in the present or future
- list of material
 - can not be empty

spring code of User

```
@Data no usages
@AllArgsConstructor
public class User {
    @NotEmpty(message = "id can not be empty")
    @Size(min = 2,message = "the id length must be greater than 2")
    private String ID;
    @NotEmpty(message = "name can not be empty")
    @Size(min = 4,max = 15,message = "the length of name must be between 4 and 15")
    private String name;
    @Email(message = "invalid email")
    private String email;

    @NotNull(message = "list cant be null")
    private ArrayList<Subject> listOfSubjects;

    @NotEmpty(message = "username can not be empty")
    @Size(min = 4,max = 20,message = "user name length must be between 4 and 20")
    private String username;

    @NotEmpty(message = "password can not be empty")
    @Pattern(regexp = "^(?=.*[A-Z])(?=.*[\\d])(?=.*[A-Za-z0-9]).+$",message = "must have special letter and capital letter")
    @Size(min = 8,max = 20,message = "password length must be between 8 and 20 ")
    private String password;

    @Pattern(regexp = "Student|Teacher")
    private String role;
}
```

VALIDATION CHECKLIST

spring code of Subject

```
@Data no usages
@AllArgsConstructor
public class Subject {

    @NotEmpty(message = "id can not be empty")
    @Size(min = 2,message = "the id length must be greater than 2")
    private String subject_ID;

    @NotEmpty(message = "name can not be empty")
    @Size(min = 10,max = 25,message = "the name length must be between 10 and 25")
    private String name;

    @NotEmpty(message = "description can not be empty")
    @Size(min = 25,message = "the description length must be greater than 25")
    private String description;
    @NotNull(message = "must assign teacher")
    private User teacher;
    💡 @NotNull(message = "students cant be null")
    private ArrayList<Student> students;
}
```

spring code of lecture

```
@Data no usages
@AllArgsConstructor
public class Lecture {

    @NotEmpty(message = "title cant be empty")
    @Size(min = 4,max = 18,message = "title length must be between 4 and 18")
    private String title;
    @NotEmpty(message = "description cant be empty")
    @Size(min = 25,message = "the description must be greater than 25")
    private String Description;

    @NotNull(message = "the duration must be detected")
    @Min(value = 30,message = "the duration must be greater than 30")
    @Max(value = 90,message = "the duration must be less than 90")
    private int duration;
    @FutureOrPresent(message = "the date must be today or in the future")
    private LocalDate date;
    @NotNull(message = "must have material")
    private ArrayList<Material> materials;
}
```